

Analyzing What Makes A Pitch Effective In Major League Baseball

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Abstract

This research attempts to predict the effectiveness of Major League Baseball (MLB) pitches using a variety of pitch characteristics. In an industry where MLB pitchers make millions of dollars per year, these results could benefit pitchers trying to improve, teams trying to effectively spend their money, or fans trying to better understand the game. Pitch-by-pitch data from the 2022 MLB season was used in logistic regression models to predict the probability of different outcomes based on pitch characteristics (e.g., speed, spin, movement, location). Separate logistic regression models were used to predict how often pitches will result in a swing and miss, solid contact (a “barrel”), or any positive result for the pitcher. These models were further split by pitcher handedness and pitch type. Backward stepwise regression was used to determine which variables to include in each model. Which variables were statistically significant, and their importance, varied from model to model, confirming our hypothesis that the impact of individual pitch characteristics differed substantially between pitch types, pitcher handedness, and batter handedness. The results of these models were used to create an interactive applet that is posted online, allowing users to explore pitch effectiveness. The applet allows a user to input pitch characteristics and to click on a location in a strike zone to predict the likelihood of different outcomes for their described pitch. The applet also includes a series of heatmaps of the likelihood of outcomes based on pitch location and a list of the MLB pitchers with the most similar pitch to the one inputted. The applet is intended to be used as an evaluation tool or to determine potential changes for a pitcher’s arsenal.

Introduction

The purpose of this project was to investigate what makes a pitch effective, specifically in Major League Baseball. More specifically, the purpose was to use pitch-by-pitch data to determine which variables were important in predicting a successful pitch and create some type of output that would succinctly display a pitch’s ability to create desired results. This research project has an emphasis on application and interpretation, so it is paramount that the output showcases the effects of changes in the different aspects of a pitch in a way that can inform decisions about pitch shape, location, and usage.

Existing Literature

Most existing research into pitch effectiveness is published in the public sphere by non-academic researchers who are baseball fans or work in the baseball industry. Many of these utilize XGboost, or some other form of complex modeling, to evaluate pitch effectiveness by creating a single statistic composed of a multitude of different pitch metrics. There are also articles on the benefits of pitch speed, spin rate, vertical approach angle, and other pitch measurements. One such article, which most closely aligns with this research project, is *The Mystic Art of Pitch Tunneling*, published in 2022 by Tieran Alexander. This article investigates the effectiveness of different pitch types, classifying pitches based on how they relate to the pitcher's fastball in terms of speed and movement. The most mainstream classification of pitch effectiveness is Stuff+, the first iteration of which was created by Eno Sarris and Max Bay at Fangraphs (Sarris, 2021). Stuff+ uses a multitude of pitch metrics, including velocity, movement, release point, and the velocity and movement differentials (against a pitcher's fastball), and grades pitchers and pitches on a scale where 100 is average (120 would be 20% better than league average and 80 would be 20% worse than league average). There is a lot of research on pitch effectiveness from an output-centric perspective, aiming to put a value on different pitches and pitchers. However, there is less public research with a direct emphasis on altering different aspects of a pitch and determining the impact of those changes. This research aims to evaluate pitching on a more straightforward probability scale while being able to articulate the effects of different pitch characteristics on the likelihood of desired outcomes.

Data Collection

This research uses multiple datasets from Baseball Savant. The data used for the modeling is 2022 pitch-by-pitch data for every pitcher with at least 1800 pitches thrown that season. The data for the models was 297,675 observations with 66 variables. The applet also utilizes seasonal pitch averages from 2021 to 2023 to define the outer bounds of the input sliders, which were also collected from Baseball Savant. The data was collected yearly, with spin data in one CSV and movement and speed information in another. When joined together, it is roughly 1800 observations with 35 variables. The data was then summarized, giving the minimum and maximum movement and spin for each combination of speed (rounded to the nearest mph), pitch type, and pitcher handedness.

Data Wrangling

New variables were computed to aid in evaluating pitch value throughout the course of this research. Some were used for other modeling approaches or in other iterations of the models, while others were input into the applet. To determine if pitchers “reached back” for better pitches and got better results at given points in a game, difference variables were created to measure the discrepancy between the speed and break of a given pitch versus other pitches of that type in that game. An overall movement variable was also created to work in conjunction with the existing variables measuring horizontal and [induced] vertical movement. Operating under the general assumption that the middle of the strike zone is the center of action, a variable was also created to represent how far from the center of the plate the pitch was located. Batter tendencies were also calculated for the three outcomes of interest. The likelihood of a batter to swing and miss, barrel, or lose run expectancy based on pitcher handedness and pitch type was appended to the data to represent the impact the batter has on outcome probabilities.

Exploratory Data Analysis

The first aspect of the project was determining which variables impacted pitch effectiveness the most. A lot of the initial research involved variable exploration and determining relationships between potential causes of effectiveness (such as speed or movement) and measures of effectiveness (such as wOBA or run value).

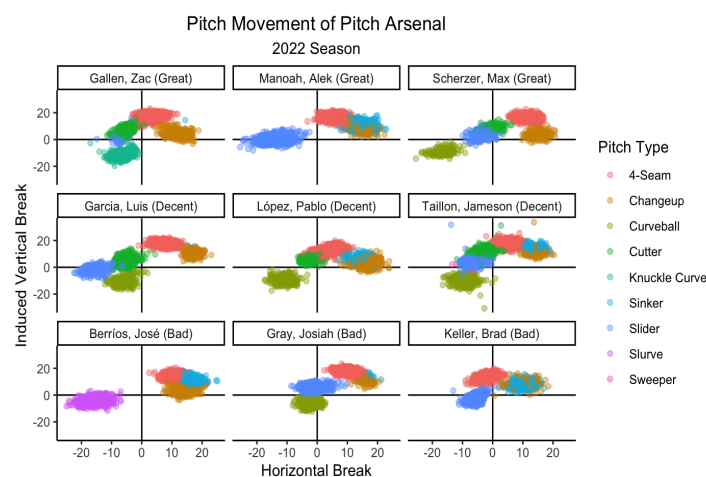


Figure 1: Movement Plots

I wanted to see if pitchers who performed with varying degrees of success had substantial

differences in their pitch shapes and arsenals, even when they had the same fastball at first glance. Nine right-handed pitchers with similar fastball speeds but different seasonal performances were compared against one another to see which aspects of their performances could be attributed to other factors. To determine which variables impacted the effectiveness of a pitch, many potential independent variables were compared against wOBA (a widely accepted measure of offensive performance). When investigating specific variables, such as the spin rate of their different pitches or batted ball type tendencies (compared against run expectancy added), there were no metrics that could principally be causing these differences. *Figure 1*, shown above, displays each pitcher's pitch movement plots. The better pitchers generally tended to have more defined, but connected pitch movement shapes; each pitch worked cohesively but had a unique purpose. However, there was not one distinct thing that good pitchers did in terms of their arsenals or pitch shapes that could be pointed to as the cause of their effectiveness. This supports the hypothesis that pitch effectiveness is a product of a multitude of different factors interacting together, including overall pitch arsenal, pitch speed, movement, location, usage, etc.

Model Testing

Multiple model types were considered for this project, weighing their predictive accuracy with complexity and reliability. Multiple linear regression was the easiest model to apply and explain, but it had poor predictive accuracy as it failed to capture important nonlinear (a one unit increase in one variable may not always have a consistent effect in the other) and interactive nuances. Random forest and GAM models were both considered and tested as well. Both modeling approaches had much better predictive accuracy (especially the random forest model). However, the goal of the project was to have a comprehensible and actionable output. GAM's smoothed splines terms and random forest's made it difficult to easily articulate the impact of changes in specific variables. Logistic regression was the modeling approach that was chosen for the project because it utilizes square terms and many variables in a model with finite coefficients, which are fairly easy to understand and apply. Furthermore, its output is in the form of log odds, which can be shifted to a probability scale, which makes a lot more sense as the output.

Logistic regression models were used to predict three response variables: swing and miss likelihood, barrel likelihood, and strike likelihood (any called strike or swing). Many input

variables were used, and paired down one at a time based on p-value. Included within the input variables were:

Candidate Variables for Logistic Regression Models	
Pitch Speed	Pitch Spin
Horizontal Movement	Vertical Movement
Horizontal Movement ²	Vertical Movement ²
Previous pitch type	Batter's seasonal average for response
Horizontal fastball pitch tunnel	Vertical fastball pitch tunnel
Horizontal movement difference vs fastball	Vertical movement difference vs fastball

Figure 2: Candidate Variables for Logistic Regression Models

For the applet, one response variable was removed, as whether the pitch was a strike was not a great measure of pitch value. The strike model was replaced with a positive run expectancy (+ RE) model as a measure of a general good outcome, predicting the likelihood that the pitch would result in an outcome with a good result for the pitcher, according to run expectancy (this would include strikes and contact that resulted in an out).

When the form of the final product was finalized, the context-dependent variables and complex interaction variables were removed from the models, as it was not realistic to expect users to know or input them into the applet. Instead, the models focused on pitch metrics and locations, which are more easily communicated and applied. The removal of context variables sacrificed some degree of model accuracy in favor of the ability to apply and comprehend the model.

Logistic Modeling

The logistic regression was done using the `glm()` function from the stats package in R. Given the importance of pitch type, pitcher handedness, and batter handedness, different models were fit for each combination of the aforementioned criteria shown below in *Figure 3* (84 total models were created):

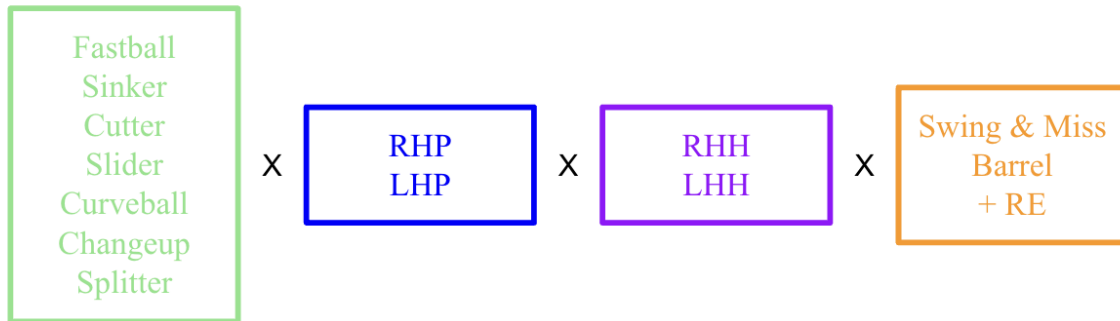


Figure 3: Logistic Model Combinations

For each model, the same input variables were tuned by using a loop to iterate through each combination of model criteria with backwards stepwise regression, which removed the least useful variable by AIC to maximize accuracy and minimize complexity. The model inputs that were fed into the backwards stepwise regression were the variables shown below in *Figure 4*:

Variables	Range / Units
Pitch Speed	60 to 100 miles per hour
Horizontal Movement	inches
Vertical Movement	inches
Total Movement	inches
Spin Rate	revolutions per minute
Average Batter Likelihood	proportion
Pitch Zone	Rectangular locations, numbered 1 - 14
Distance From Center of Strike Zone	Proportion from center to edge of strike zone
Horizontal Location & Horizontal Location ²	-12.75 to 12.75 inches
Vertical Location & Vertical Location ²	Proportion from center to edge of strike zone

Figure 4: Logistic Model Inputs

Each model was saved as a collection of coefficients, but can also be put into a more traditional model form (the example below predicts swing and miss for sliders by a right-handed pitcher to a right-handed hitter):

$$\log odds = -6.76 + 0.03(pfx_x) + 0.03(pfx_z) - 0.04(pfx_total) - 0.46(dist_x) - 0.83(dist_x^2) - 0.73(dist_z) - 0.96(dist_z^2) + 2.07(dist_total) + 0.0004(spin) + 18.21(batter_pred) \pm \beta(zone\ 1 \dots 14)$$

Each value is input into the model and multiplied against its coefficient. For example, in the model shown above, every extra rpm (holding all else constant) increased the log odds prediction of swing and miss by 0.0004.

Each pitch zone is a separate indicator variable that says whether or not the pitch is located within that zone. The batter_pred variable accounts for the strength of the hitter in a prediction and for the interactive applet, the average batter likelihood by pitch type and matchup handedness is inputted in the model since we wanted to describe general trends and not pitch effectiveness against specific hitters. Overall, the variables are meant to represent the pitch's speed and shape, as well as location. The pitch's zone was used as a measure of general location, with the understanding that the rough location of a pitch also influences the outcome, in addition to minute differences in pitch location. The square terms for location and the variable measuring the pitch location's distance from the center of the strike zone were included to account for the generally high offensive output recorded from pitches that are near the sweet spot of the bat.

Shiny App

The deliverable for this research project is an R Shiny App, which is posted online for others to use (evanwu.shinyapps.io/Pitch_Outcome_Model/). The app takes eight user inputs and

outputs three graphs and a table. The user inputs matchup information, pitch type, pitch metrics, and location, and outputs the probability that the resulting pitch will cause a swing and miss, be hit at optimal launch angle and velocity, or result in a good outcome for the pitcher.

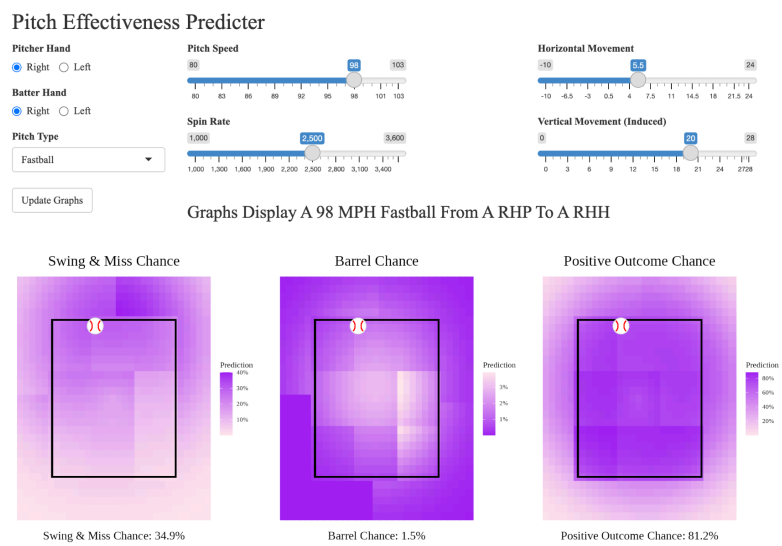


Figure 5: Shiny App Inputs & Heat Maps

These heat maps show the likelihood of each of the three response outcomes for every pitch location, which is displayed above in Figure 5. Below the heat maps is a pitch comparisons table, which provides the ten pitchers whose average pitch from the 2024 MLB season is most like the user-inputted values, a snippet of which is shown below in Figure 6. In conjunction with the comparison pitch’s metrics, the table also displays its performance in the three outcome variables, as well as a similarity score. Similarity is based on a z-score metric where pitches that have lower scores are closer to the input values.

Pitch	Name	Speed	Spin Rate	Hor Break	IVB	Swing&Miss%	Barrel%	+RE%	similarity
Slider	Jorge Alcala	89.9	2,504	-3.56	4.35	13.6%	0.8%	53.3%	0.62
Cutter	Zack Kelly	91.1	2,486	-2.22	6.81	8.6%	1.4%	60.8%	0.64
Cutter	Frankie Montas	90.4	2,427	-1.94	5.78	9.1%	2.6%	54.1%	0.70
Slider	Carson Fulmer	88.0	2,493	-4.25	5.27	11.8%	1.7%	54.4%	0.71

Figure 6: Shiny App Pitch Comparisons Table

The app utilizes seasonal pitch metric averages of every different pitcher from the 2021-2023 MLB seasons to determine the lower and upper bounds of the input sliders (based on handedness and pitch type as well), and uses seasonal averages from the 2024 season for the pitch comparisons table. The heat maps are constructed using the logistic regression models' coefficients, as well as predictions based on the pitch zone. The zone represents the general pitch location and is accounted for using a CSV that has a prediction value for each zone and model. Each heat map is constructed using the ggplot2 package in R, using a 20x20 grid of predictions based on pitch location, with each point representing 0.25% of the strike zone.

The heat maps are colored using a color scale from pink to purple, with purple representing the values that are most desirable to the pitcher. Higher values for swing and miss and positive run expectancy are purple because those are good outcomes for the pitcher, while lower values of barrel rate are pink, as the pitcher wants to avoid solid contact.

The app also outputs predictions for specific locations, which can be input by the user by clicking on any of the heat maps in the desired location. The app defaults to a pitch in the center of the strike zone and displays a small baseball graphic, which will move to denote whichever pitch location is selected. The predicted outcome likelihoods for the selected pitch location are displayed at the bottom of each graph.

The app updates based on pitch location, but does not update when the other input metrics are changed until the "Update Graphs" button is clicked. This was done to prevent the app from continuously iterating through input metrics as the input sliders were changed, which was causing substantial buffering and computational processing power issues. As a result of this change, the graphs do not initially display when the app loads (which was not the original

intention), so the user must click the update button in order to see the graphs and update them to match the inputs if they have changed them. Since the app updates upon clicking the button, a text callout informs the user if their current inputs do not match the output, so that they can update the graphs when the inputs are set to their liking.

Conclusion

Using logistic regression on pitch-by-pitch data, this app predicts the likelihood of swing and misses, solid contact, and positive outcomes. Through the online applet, users are able to explore how different pitch characteristics impact the likelihood of those outcomes. The focus on interpretation over complexity allows users to take what they learn and apply it to make effective changes. This applet is intended to be used as a tool and quick reference for people with varying degrees of baseball statistics/data knowledge, so that all users can benefit from the insights this applet provides.

Limitations

This analysis uses data from MLB, which limits its ability to be accurately applied to performance at other levels of play. There is also limited data for certain pitch types and matchups, making it difficult to draw strong conclusions for those scenarios. The dataset is from the 2022 season, which may not fully reflect recent pitch shape trends or other factors, such as the pitch clock.

The focus of this project was interpretability, which caused us to ignore certain complexities in the modeling process, such as interactions between variables and other nuanced inputs, such as the previous pitch thrown, a pitcher's arsenal information, etc. The purpose of the

app was to focus on interpretation, which is why these inputs were omitted in the final iteration of the models, but these variables would likely add more predictive accuracy if the purpose was solely to predict outcomes as accurately as possible.

Future Directions

This is an app intended for pitch location decisions and not for pitch design. In future iterations, incorporating the pitcher's arsenal (which includes the pitches the pitcher throws, and their shape/speed) would allow for some very interesting insights. Using this information to explore pitch sequencing, usage proportions, and design (modifying pitches to an optimal shape) would provide more robust and complex insights to pitchers. While these may be too complex to input or output on an app, they would be invaluable tools for pitchers to use. Baseball Savant also recently released arm angle data for the 2024 season. Models that take into account how the pitch is released, such as arm angle, release point, and vertical approach angle, would provide insights into where a pitch could be thrown most effectively and how it is likely to perform. Similar to the pitch arsenal information, this is likely too convoluted to be an aspect of an app such as this, but it would be interesting to see, and could possibly be included in a tool that is intended to intake and output more specific and distinct information.

Appendix

Software: R

R Packages: broom, concaveman, forcats, ggforce, ggimage, glmnet, gt, knitr, Metrics, olsrr, patchwork, RColorBrewer, ranger, readxl, scales, shiny, shinyjs, stringr, tidyverse

Data

Pitch-by-Pitch Data 2022:

baseballsavant.mlb.com/statcast_search?hfPT=&hfAB=&hfGT=R%7C&hfPR=&hfZ=&hfStadium=&hfBBL=&hfNewZones=&hfPull=&hfC=&hfSea=2022%7C&hfSit=&player_type=pitcher&hfOuts=&hfOpponent=&pitcher_throws=&batter_stands=&hfSA=&game_date_gt=&game_date_lt=&hfMo=&hfTeam=&home_road=&hfRO=&position=&hfInfield=&hfOutfield=&hfInn=&hfBBT=&hfFlag=&metric_1=&group_by=name&min_pitches=1750&min_results=0&min_pas=0&sort_col=pitches&player_event_sort=api_p_release_speed&sort_order=desc#results

Pitch-by-Pitch Data 2024:

baseballsavant.mlb.com/statcast_search?hfPT=&hfAB=&hfGT=R%7C&hfPR=&hfZ=&hfStadium=&hfBBL=&hfNewZones=&hfPull=&hfC=&hfSea=2024%7C&hfSit=&player_type=pitcher&hfOuts=&hfOpponent=&pitcher_throws=&batter_stands=&hfSA=&game_date_gt=&game_date_lt=&hfMo=&hfTeam=&home_road=&hfRO=&position=&hfInfield=&hfOutfield=&hfInn=&hfBBT=&hfFlag=&metric_1=&group_by=team&min_pitches=0&min_results=0&min_pas=0&sort_col=pitches&player_event_sort=api_p_release_speed&sort_order=desc#results

Seasonal Pitch Averages for Spin Rate & Movement (for years 2021 - 2023):

https://baseballsavant.mlb.com/leaderboard/pitch-arsenals?year=2021&min=100&type=avg_spin&hand=

https://baseballsavant.mlb.com/leaderboard/pitch-movement?year=2021&min=100&pitch_type=ALL&hand=

References

Alexander, T. (2022, August 23). *The mystic art of Pitch Tunneling*. Prospects Live.

<https://www.prospectslive.com/the-mystic-art-of-pitch-tunneling/>

Sarris, E. (2021, June 11). *The pitcher report: What exactly is stuff+? featuring Rich Hill, Sam Long, and more*. The New York Times.

<https://www.nytimes.com/athletic/2641834/2021/06/11/the-pitcher-report-what-exactly-is-stuff-featuring-rich-hill-sam-long-and-more/>